

Data Flow Diagram

Date	4 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

External Entities (Ovals)

Entity	Description
Loan Applicant	A person who submits personal and financial details through the Smart Lender web form to check loan eligibility
Bank Analyst	A bank professional who uses the system to evaluate multiple loan applications quickly during peak periods

Processes (Numbered Rectangles)

#	Name	What It Does
1	Collect & Validate Input	Receives applicant details from the web form and validates all 11 input fields before processing
2	Preprocess Data	Handles missing values using mean/mode imputation and encodes categorical features using LabelEncoder
3	Train & Evaluate ML Models	Trains Decision Tree, Random Forest, KNN, and XGBoost models and selects the best one
4	Run Prediction (Flask /predict)	Loads model.pkl and runs prediction on the encoded applicant feature array
5	Display Result	Renders result.html showing Approved/Rejected badge with applicant summary

Data Stores (Solid Rectangles)

Data Store	Description
Loan Dataset (loan.csv)	Historical loan records with applicant details used for training all four ML models
Trained Model (model.pkl)	Best performing XGBoost model saved using pickle after training - loaded at Flask app startup
Prediction Log	Stores all prediction results for audit, compliance review, and future model retraining

DFD - Smart Lender: Loan Approval Prediction

